

ProcuraAI

Autonomous Procurement Intelligence

ROCKATHON'26 — Problem Statement 01: Autonomous Commerce Engineering

Team

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"From buying intent to authorized action — autonomously, transparently, and responsibly."

Procurement Is Still a Human Bottleneck

The traditional procurement workflow remains fragmented, opaque, and slow — even for routine purchases.

01

Business Requirement Defined

Stakeholder articulates need in natural language or informal brief.

02

Manual Vendor Discovery

Cross-source search with no systematic aggregation or normalization.

03

Time-Consuming Quote Comparison

Manual side-by-side evaluation with no structured scoring model.

04

Constraint Validation & Conflicts

Budget, spec, and delivery conflicts resolved informally — if at all.

01

Budget & Authorization Checks

Manual policy lookups with no enforcement layer; easy to bypass.

02

Approval Bottlenecks & Delays

Sequential sign-offs add days of latency to routine decisions.

03

Order Placement & Documentation

Manual data entry into systems with fragmented record-keeping.

04

Fragmented Decision Traceability

No unified audit trail; decisions driven by availability, not optimization.

RESULT • Days to weeks for routine purchases • no clear audit trail • decisions driven by availability, not optimization.

The Real Problem Isn't Finding a Product

It is making the right purchasing decision under constraints.

Traditional System Asks

"Which products match the specs?"
Product discovery only. No constraint enforcement, no risk assessment, no policy layer, no justification.

ProcuraAI Asks

"Which purchase best satisfies requirements, minimizes risk, stays within authorization, and **can be justified?**"
Decision intelligence across all dimensions — simultaneously and autonomously.

 **Example Brief:** "Buy 10 laptops under ₹45,000 each, with ≥16GB RAM, ≥512GB SSD and delivery within 7 days." — This single sentence encodes hard constraints, soft preferences, and budget policy simultaneously.



UNDERSTAND

Extract intent and constraints from natural language



REASON

Evaluate alternatives using multi-dimensional criteria



CONTROL

Enforce authorization boundaries and policy

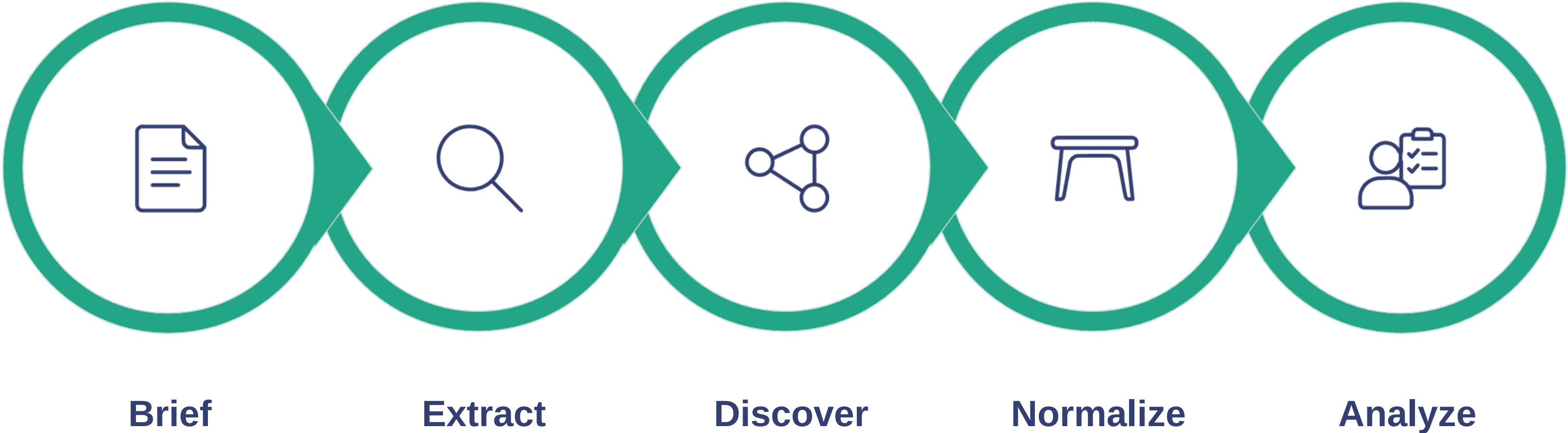


EXPLAIN

Justify every decision with evidence

Meet ProcuraAI

An autonomous, policy-aware procurement agent that converts natural-language buying requirements into structured procurement decisions and executes them within defined boundaries.



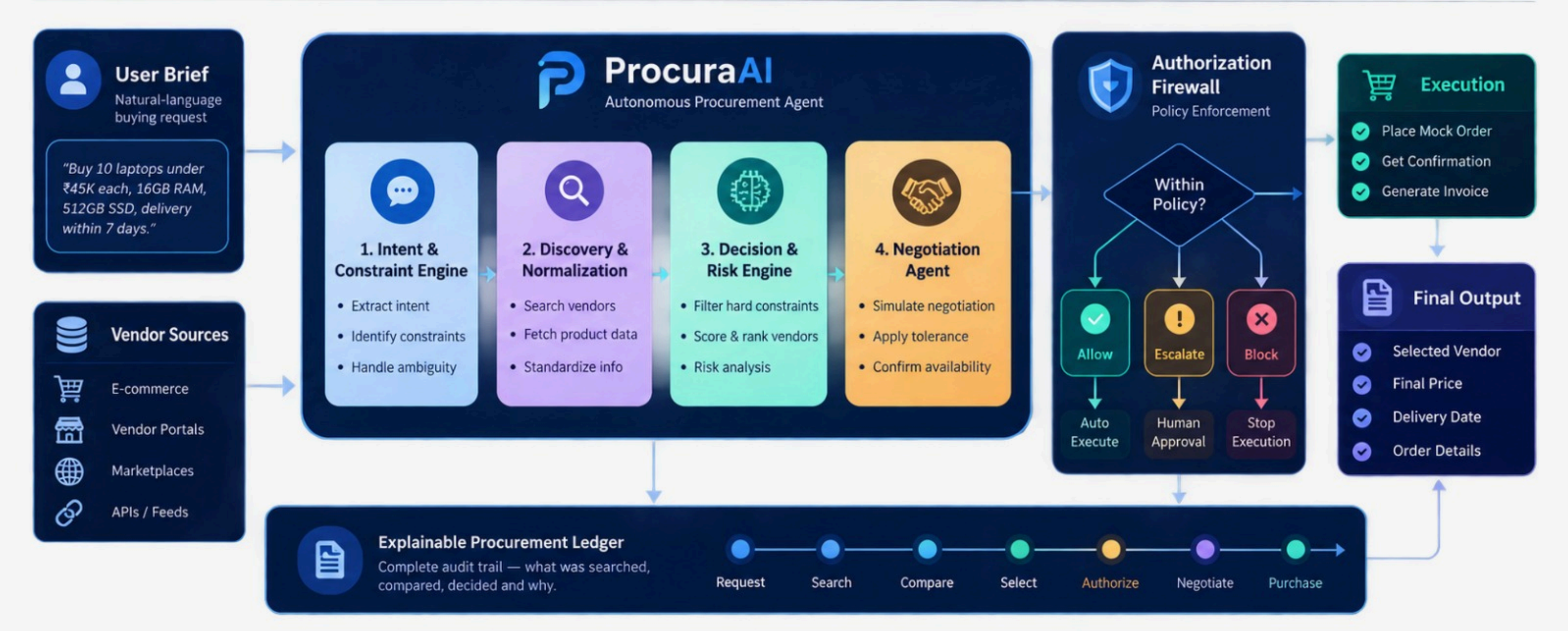
ProcuraAI does not merely search for products — it executes a complete decision lifecycle, ensuring every purchase is constraint-compliant, policy-authorized, and fully explainable.

Core Principle

Autonomy within boundaries. The agent acts decisively on routine purchases, escalates intelligently when authorization is required, and never exceeds defined policy limits.

ProcuraAI System Architecture

A complete autonomous procurement workflow with policy-controlled execution.



Constraint Intelligence Engine

The first innovation pillar: converting natural language into machine-checkable constraints — the foundation of every downstream decision.

📄 **Example Buying Brief:** "Buy 10 laptops under ₹45,000 each, with 16GB RAM, 512GB SSD and delivery within 7 days."

Hard Constraints

- Quantity = 10
- Budget $\leq$ ₹45,000 per unit
- RAM $\geq$ 16GB
- SSD $\geq$ 512GB
- Delivery $\leq$ 7 days

Non-negotiable. Any vendor failing a hard constraint is automatically eliminated.

Soft Preferences

- Brand preference
- Warranty coverage
- Return policy quality

Influence the ranking score. A vendor without preferred brand is not eliminated but ranked lower.

Ambiguous / Unclear

- "Good performance" → Requires clarification
- "Affordable" → Budget threshold undefined

Agent proactively asks: *"What is your maximum budget per laptop?"* before proceeding.

Ambiguity Handling

When input is underspecified, the agent does not guess. It identifies the ambiguous field, pauses decision execution, and requests targeted clarification from the requester before continuing.

Constraint Relaxation

When no vendor satisfies all hard constraints simultaneously, the engine performs a relaxation analysis — showing the user exactly which constraint could be softened, and the downstream consequence of each relaxation option.

Risk-Aware Decision Engine

Cheapest ≠ Best. Hard constraints are applied first to eliminate non-compliant vendors. Remaining options are ranked using a multi-objective scoring model that captures total procurement outcome — not just unit price.

Vendor	Price / Unit	Delivery	Reliability	Risk Level	Score
Vendor A	₹42,000	5 days	94%	Low	91.4
Vendor B	₹39,000	7 days	63%	High	71.8
Vendor C	₹44,000	4 days	91%	Low	89.7

Decision Factors

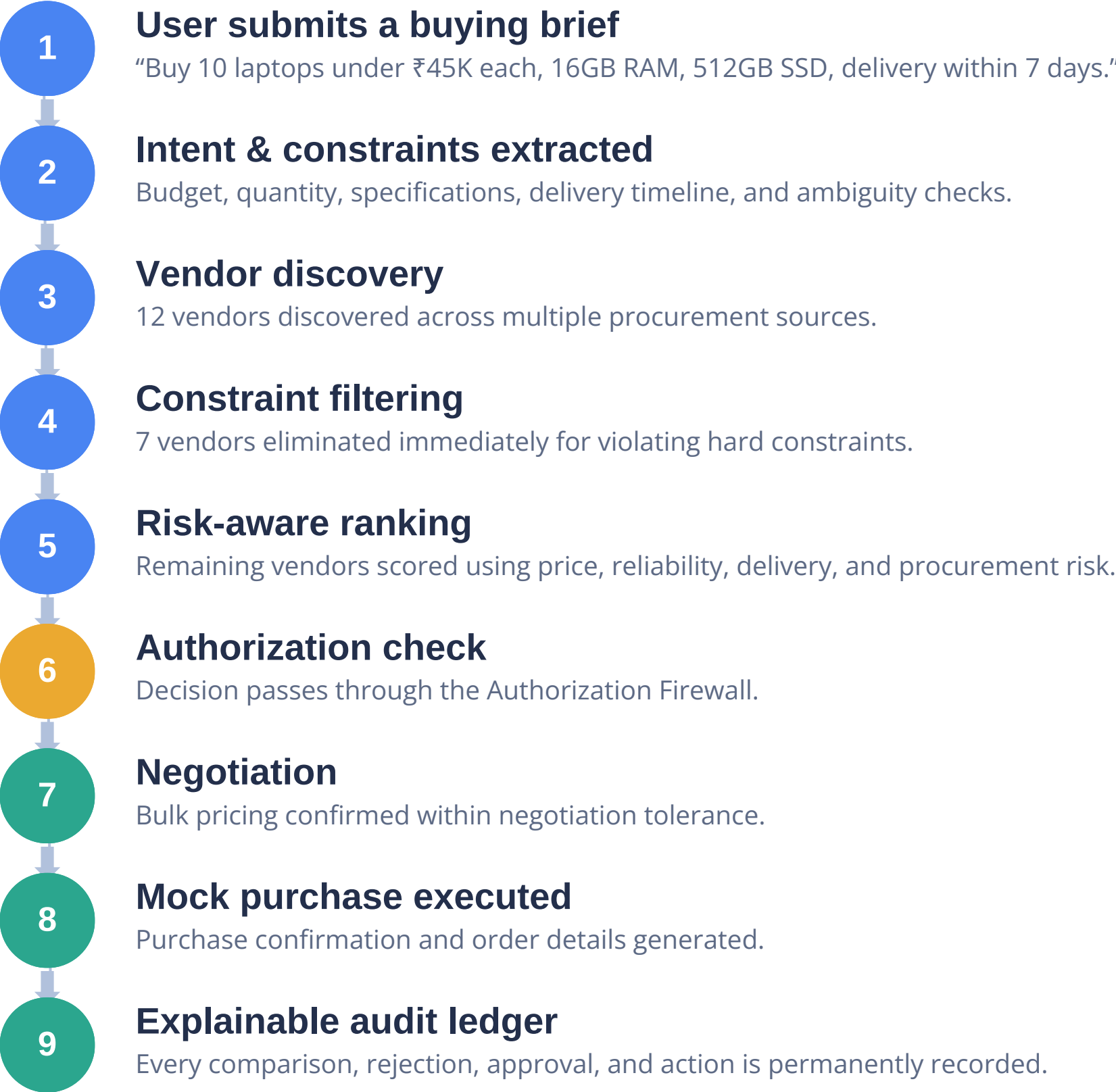
- **Constraint satisfaction**
Hard constraint pass/fail gate applied before scoring
- **Value & delivery speed**
Price and timeline jointly optimized
- **Reliability & risk**
Vendor track record and procurement risk band

Outcome: Vendor A Selected

Vendor A meets all hard constraints, achieves the highest reliability score (94%), carries the lowest procurement risk, and delivers the best overall outcome — despite not being the cheapest option. Vendor B's low reliability (63%) and high risk profile disqualify it from serious consideration regardless of price advantage.

One Request. One Intelligent Procurement Journey.

From a natural-language buying brief to an explainable, policy-compliant purchase.



PROCESS

CONTROL

EXECUTE

DECISION HIGHLIGHT

ProcuraAI's Final Decision

Example procurement outcome

LOW RISK

SELECTED VENDOR

Vendor A

TOTAL COST

₹4.2L

DELIVERY

5 Days

RELIABILITY

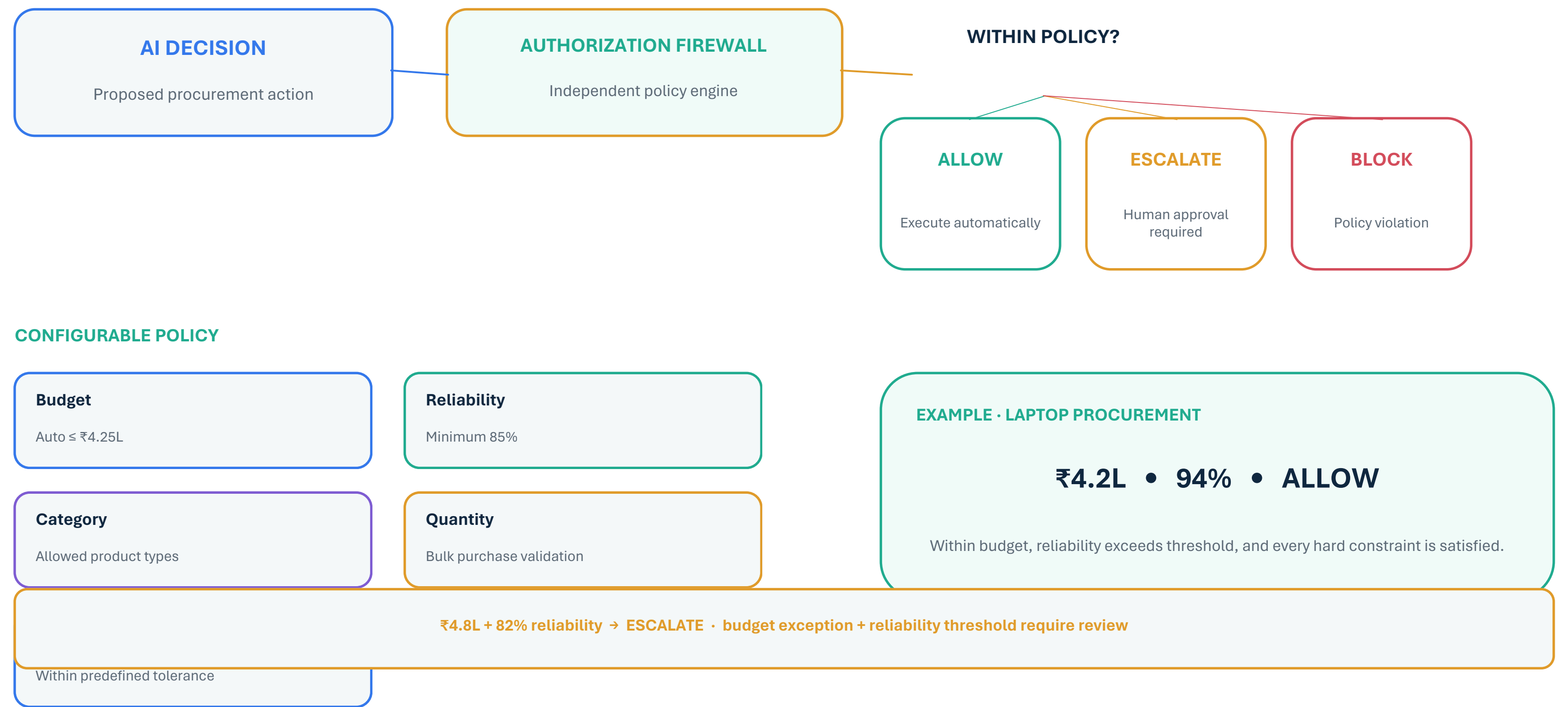
94%

DECISION RATIONALE

Vendor A is selected because it satisfies all hard constraints, remains within policy, delivers in 5 days, and achieves the highest overall procurement score.

The Agent Can Act — But Never Beyond Its Authority

A hard policy enforcement layer validates every procurement decision before execution.



Decision-Centric Human Oversight

Do not ask humans to approve every routine action. **Escalate only when it matters.** Human attention is a scarce resource — the agent preserves it for genuinely consequential decisions.

Escalation Triggers

Authorization Exception

Total cost exceeds autonomous execution limit

Critical Ambiguity

Requirements too underspecified to proceed safely

High-Risk Decision

Best available vendor has low reliability score

Low Confidence

Multiple equally-ranked options with no clear winner

Constraint Conflict

No fully compliant option exists across all hard constraints

Decision Packet Delivered to Approver

Proposed Purchase

Vendor, quantity, unit price, total cost, authorized amount, and variance

Escalation Context

Reason for escalation, risk level, delivery implications

Alternatives with Trade-offs

All compliant alternatives with ranked comparison

Agent Recommendation

Preferred action with full justification

Human Actions Available

APPROVE

Accept recommendation

REJECT

Decline, request alternatives

MODIFY

Adjust constraints, recalculate

Why-Not Engine

Every decision has two sides: **what was chosen** and **why alternatives were rejected**. The Why-Not Engine generates structured rejection rationale for every eliminated vendor — making the agent's logic fully auditable.

✓ **SELECTED** — Vendor A

- Meets all mandatory hard constraints
- Within budget: ₹42K ≤ ₹45K limit
- On time: 5 days ≤ 7-day requirement
- High reliability: 94%
- Low procurement risk classification

✗ **REJECTED** — Vendor B

- Reliability critically low: 63% (far below acceptable threshold)
- High-risk procurement classification
- Delivery at maximum boundary with insufficient reliability buffer

✗ **REJECTED** — Vendor C

- Total order cost: ₹44K × 10 = ₹4.4L
- Exceeds authorization limit at policy boundary
- Requires escalation — not auto-executable

✗ **REJECTED** — Vendor D

- Reliability 78% — below minimum policy threshold of 85%
- Policy violation: mandatory reliability floor not met

Core Principle

The agent explains both its choice and its rejections. A decision that cannot be justified should not be executed. The Why-Not Engine is what separates a procurement intelligence system from a recommendation engine.

Negotiation + Failure Recovery

Autonomy doesn't stop when the first answer fails. **The agent handles exceptions gracefully — not with error states.** Failure recovery and negotiation are first-class capabilities, not afterthoughts.

Scenario 1

Vendor Unavailable

- Top-ranked vendor reports unavailability
- Agent re-checks next compliant alternative automatically
- Recalculates risk and delivery impact for new selection
- Proceeds without human interruption if within policy

Scenario 2

No Exact Match

- No vendor satisfies all hard constraints simultaneously
- Agent performs constraint relaxation analysis
- Presents ranked trade-off options per relaxation path
- Asks human: *which constraint may be relaxed?*

Scenario 3

Negotiation Simulation

- Vendor quote received after initial selection
- Agent applies pre-defined negotiation policy and tolerance thresholds
- Within tolerance → accept automatically
- Outside tolerance → escalate for human decision

Design Philosophy

Every failure path has a defined recovery strategy. The agent never reaches a dead end — it either resolves the exception autonomously, or escalates with a structured decision packet. **Graceful recovery and escalation, not error states.**

Explainable Procurement Ledger

Every action leaves evidence. **Complete, immutable audit trail from request to execution** — answering every question a stakeholder, auditor, or judge might ask.



Request

Search

Compare

Decide

Authorize

The Ledger Answers Three Questions

"Why this?"

Why was the selected vendor chosen? Full scoring rationale, constraint satisfaction matrix, and comparative ranking evidence.

"Why not that?"

Why were other vendors rejected? Specific failure reason per vendor — hard constraint violation, policy threshold breach, or risk classification.

"What happened?"

What actions did the agent perform? Complete step-by-step execution log from natural language input to purchase confirmation.

Five Innovation Pillars

01

Constraint Intelligence Engine

Natural language → machine-checkable constraints

02

Risk-Aware Decision Engine

Multi-objective scoring beyond price

03

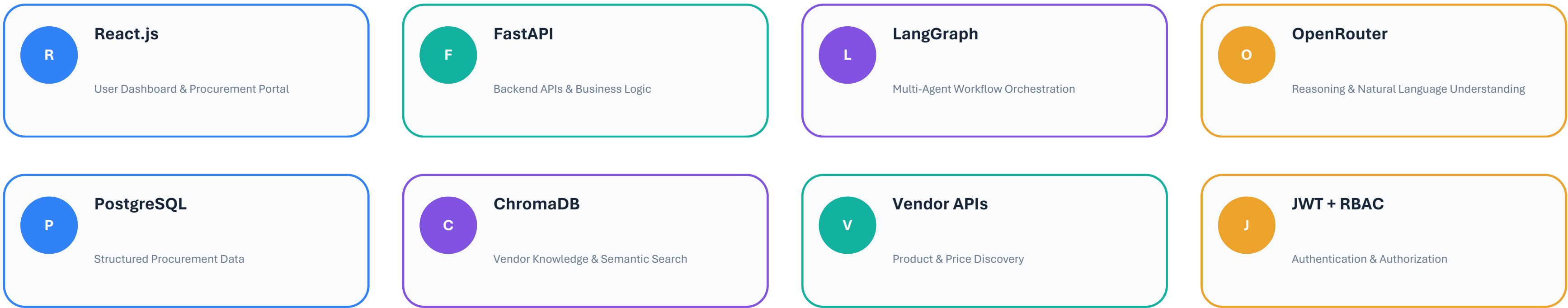
Why-Not Engine

Structured rejection rationale for every alternative

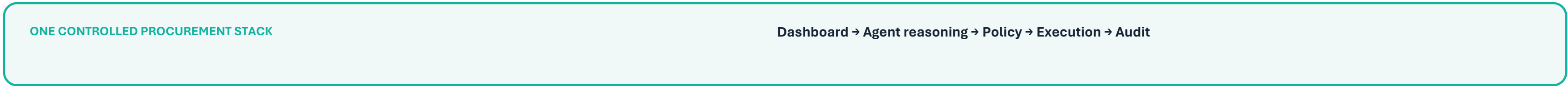
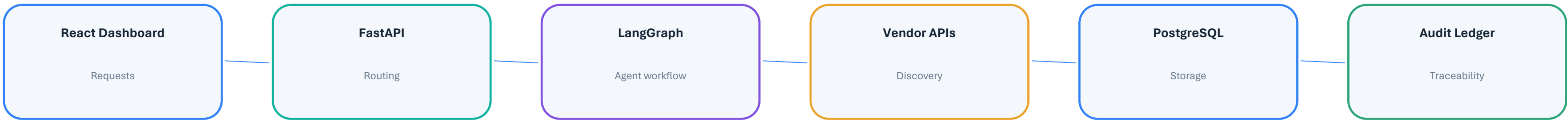
Technology Stack

Enterprise-ready technologies powering ProcuraAI’s autonomous procurement workflow.

CORE TECHNOLOGIES



HOW EACH TECHNOLOGY FITS PROCURA AI



INTEGRATION PATH

React → FastAPI → LangGraph → Vendor APIs → PostgreSQL → Audit Ledger